Task 10

Task 1:

Initial:

- front = 0
- rear = 0
- count = 0
- array = [_ _ _ _ _]

1. enQueue(10)

- Insert at rear = 0
- rear = (0+1)%5 = 1
- count = 1

Array: [10 _ _ _ _]
front = 0, rear = 1

2. enQueue(20)

- Insert at rear = 1
- rear = 2
- count = 2

Array: [10 20 _ _ _]
front = 0, rear = 2

3. enQueue(30)

- Insert at rear = 2
- rear = 3
- count = 3

Array: [10 20 30 _ _]
front = 0, rear = 3

4. deQueue()

- Remove from front = 0 (value = 10)
- front = 1
- count = 2

Array: [_ 20 30 _ _]
front = 1, rear = 3

5. enQueue(40)

- Insert at rear = 3
- rear = 4
- count = 3

Array: [_ 20 30 40 _]
front = 1, rear = 4

6. enQueue(50)

- Insert at rear = 4
- rear = (4+1)%5 = 0
- count = 4

Array: [_ 20 30 40 50]
front = 1, rear = 0

7. enQueue(60)

- Insert at rear = 0
- rear = 1
- count = 5 (FULL)

Array: [60 20 30 40 50]
front = 1, rear = 1

- **Front value = 20**
- **Rear position = 1**

Task 2:

Queue Full Condition

When the count == SIZE, it means queue is full when number of elements becomes equal to size and no more space left to insert new elements. Hence it means when all positions are filled, queue is full.

Queue Empty Condition

When the count == 0 , it means that queue is empty and there are no elements in it.And there is nothing to delete.Hence it means when there is no element in queue, it is empty.